

Evaluating the Second Law of Infodynamics Against Established Physics: A Critical Assessment with Falsifiable Predictions

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Abstract

Vopson’s Second Law of Infodynamics (2022) posits that the information entropy of physical systems tends to decrease or remain constant over time. While the empirical pattern has been documented across genetic, atomic, and cosmological systems, the law has remained a postulate lacking derivation from established physics. This paper evaluates the law against quantum chromodynamics, the Newtonian limit of general relativity with a QCD-derived vacuum–matter coupling, and algorithmic information theory.

We introduce *effective complexity* (Gell-Mann & Lloyd, 2003) as the central quantity and show that, under a specific physical condition—the existence of a vacuum–matter coupling $\alpha > 0$ derivable from the QCD sigma terms of the nucleon—the effective complexity of the matter density field increases monotonically toward a unique fixed point. The derivation proceeds through four self-contained steps: (1) the in-medium chiral condensate shift at finite baryon density, a standard result of nuclear physics, identifies $\alpha_{\text{bare}} = (\sigma_\pi + \sigma_s)/m_N \approx 0.096$; (2) reduction by the baryon fraction and nonlinear self-screening (verified in N-body simulations presented here) gives $\alpha_{\text{predicted}} = 0.005$, in agreement with the observed value $\alpha = 0.003$ within a factor of 1.7; (3) the cyclical operator Φ_{cycle} (vacuum assignment $\rightarrow$ metric solution $\rightarrow$ matter redistribution) satisfies the Banach contraction mapping theorem under Condition C (numerically verified); (4) a monotonicity lemma links energy decrease to complexity growth along the Picard iterates.

The cosmic web serves as the decisive empirical arena: its supra-horizon coherence, topological universality (Vazza & Feletti, 2020), and rapid formation at $z > 10$ are explained as consequences of the unique fixed point—features that Λ CDM explains statistically but not individually. We define two complementary information bits: the *infodynamic bit* (Vopson, thermodynamic) and the *QCD bit*

(quantum-thermodynamic). We introduce *simulability* $\mathcal{S} = 1 - k$ as a continuous epistemological criterion for physical theories, standing alongside Popper's falsifiability. Eight falsifiable predictions are presented, distinguishable from Λ CDM with DESI, Euclid, SKA, and JWST data.

Keywords: infodynamics, effective complexity, second law, gravitating vacuum, cosmic web, structure formation, information physics, simulability, Kolmogorov complexity, QCD sigma terms, chiral condensate, contraction mapping, Plateau's laws, falsifiability

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1 Introduction

In 2019, Vopson proposed the mass–energy–information equivalence principle, suggesting that information is not merely physical but possesses a finite, quantifiable mass [Vopson, 2019]. In 2022, Vopson and Lepadatu formulated the Second Law of Infodynamics: the information entropy of physical systems tends to decrease or remain constant over time [Vopson and Lepadatu, 2022]. In 2023, Vopson extended these results to argue that the observed tendency toward information optimization supports the simulated universe hypothesis [Vopson, 2023].

The empirical pattern is real. Systems from genetic codes to atomic symmetries to cosmic structures exhibit decreasing information entropy [Vopson, 2023]. The theoretical grounding, however, has been challenged on multiple fronts: the mass–energy–information equivalence lacks empirical support and leads to paradoxes when applied to different physical representations of the same information [Burgin and Mikkilineni, 2022, Lairez, 2024]; the conflation of Shannon entropy with thermodynamic entropy introduces conceptual confusion [Hossenfelder, 2025]; and the simulation hypothesis, while provocative, is widely regarded as unfalsifiable.

This paper evaluates the Second Law of Infodynamics against established physics. We ask: can the law be derived—not postulated—from quantum chromodynamics, general relativity, and algorithmic information theory? And if so, what does it predict that the standard cosmological model does not?

The physical mechanism under examination is the gravitating vacuum framework [Kriger, 2026a,b], which separates the cosmological constant Λ (a geometric property of space-time) from the quantum vacuum energy ρ_{vac} (a field-theoretic quantity). This paper is *self-contained*: every derivation is presented in full, including the QCD calculation of the coupling constant (§5), the N-body simulation results (§7), and the contraction mapping analysis (§10). No result is assumed from external unpublished work without being rederived here.

The *decisive empirical arena* is the cosmic web. The large-scale structure of the universe—voids, walls, filaments, clusters—is the most complex structure in the observable universe. It presents three features that Λ CDM explains statistically but not individually: supra-horizon coherence of filaments extending over 400–500 Mpc [Gott et al., 2005], topological universality shared with neuronal networks across 10^{27} orders of magnitude in scale [Vazza and Feletti, 2020], and rapid formation of massive structures at $z > 10$ observed by JWST [Labbé et al., 2023]. We examine whether these features can be understood as consequences of a single mathematical structure.

Notation. Throughout, we use natural units where $c = 1$ except in equations where c appears explicitly for clarity. G is Newton’s gravitational constant, k_B is Boltzmann’s constant, $\hbar = 1$, and metric signature $(-, +, +, +)$.

2 Logical Structure of the Argument

The argument proceeds through the following chain, with each step self-contained:

1. **Definitional foundation** (§4): The identity *structure* $\equiv$ *compressible information* $\equiv$ *effective complexity* K_{eff} is established through Kolmogorov complexity and the Gell-Mann–Lloyd decomposition.
2. **QCD derivation of α** (§5): The vacuum–matter coupling is derived from the in-medium chiral condensate shift, the Gell-Mann–Oakes–Renner relation, and the experimentally measured sigma terms. The derivation is presented in full, including the equation-of-state argument, the trace anomaly consistency check, and the role of the strangeness sigma term.
3. **Modified gravitational dynamics** (§6): The modified Poisson equation and two-phase dynamics (repulsive voids, attractive bound structures) are derived from the ansatz.
4. **N-body simulations** (§7): The particle-mesh algorithm, initial conditions, convergence study, and the discovery of nonlinear self-screening are presented. The screening factor of ~ 3 is established.
5. **Two information bits** (§8): The infodynamic bit (Vopson, thermodynamic) and the QCD bit (quantum-thermodynamic) are defined as complementary measures.
6. **Simulability** (§9): Simulability $\mathcal{S} = 1 - k$ is introduced as a continuous epistemological criterion.
7. **Variational principle and contraction mapping** (§10): The cosmic web functional, the cyclical operator, and the contraction property are derived with explicit Lipschitz estimates for each operator. The contraction is established rigorously in the linear regime and verified numerically in the nonlinear regime.
8. **The Second Law as a conditional theorem** (§11): Under Condition C (numerically verified contraction), the monotonicity of K_{eff} is proved via a lemma linking energy decrease to complexity growth.
9. **Predictions** (§12): Eight falsifiable predictions, distinguishable from Λ CDM.

The paper explicitly distinguishes between what is *proved* (in the mathematical sense, given stated axioms), what is *derived* (from established physics with stated assumptions), and what is *conjectured* (plausible but unproven). This distinction is maintained throughout.

3 Background and Related Work

3.1 Information physics

The idea that information plays a fundamental role in physics has a long history. Wheeler’s “it from bit” programme [Wheeler, 1990] proposed that physical reality arises from information-theoretic processes. Landauer’s principle [Landauer, 1961] established

that logical irreversibility implies physical irreversibility, with experimental verification by Bérut et al. [Bérut et al., 2012]. Bekenstein [Bekenstein, 1973] showed that black hole entropy is proportional to horizon area. The holographic principle [’t Hooft, 1993, Susskind, 1995] extended this to arbitrary regions.

Verlinde’s entropic gravity [Verlinde, 2011] proposed that gravitational attraction arises from entropy maximization on holographic screens. While controversial, it demonstrated that gravitational dynamics can be reformulated in information-theoretic terms.

3.2 Vopson’s programme

Vopson [Vopson, 2019] proposed the mass–energy–information (M/E/I) equivalence principle: a bit of information at thermal equilibrium has rest mass $m_{\text{bit}} = k_B T \ln 2 / c^2$. In Vopson and Lepadatu [2022], the Second Law of Infodynamics was formulated from micromagnetic Monte Carlo simulations. In Vopson [2023], the observation was extended to genetic, atomic, and cosmological systems.

Critiques have focused on: (a) absence of empirical verification for the M/E/I principle [Burgin and Mikkilineni, 2022, Lairez, 2024]; (b) conflation of Shannon and thermodynamic entropy [Hossenfelder, 2025]; (c) unfalsifiability of the simulation interpretation. We address each by providing rigorous definitions and derivations.

3.3 Effective complexity

Gell-Mann and Lloyd [Gell-Mann and Lloyd, 2003] introduced *effective complexity* as the length of a concise description of a system’s regularities, distinguishing it from both Kolmogorov complexity (total information) and Shannon entropy (average information per symbol). This decomposition resolves the conflation between different notions of “information” in Vopson’s programme.

3.4 The cosmic web

The large-scale structure was first mapped in the CfA survey [de Lapparent et al., 1986]. N-body simulations within Λ CDM reproduce its statistical properties [Springel et al., 2005]. Voids occupy approximately 60% of the cosmic volume [Pan et al., 2012], with a universal density profile [Hamaus et al., 2014]. JWST has revealed filamentary structures at $z > 6$ [Wang et al., 2023]. The cosmic web–neuronal network comparison by Vazza and Feletti [Vazza and Feletti, 2020] documented quantitative topological identity across 10^{27} orders of magnitude in scale.

A note on what Λ CDM explains. The standard model predicts the *statistical ensemble* of cosmic web configurations (power spectrum shape, halo mass function, void size distribution) through the inflationary power spectrum. Long filaments arise naturally from this spectrum. What Λ CDM does not predict is any *specific* realization—the individual coherence of a particular filament across 500 Mpc is a contingent outcome of the random seed, not a prediction of the theory. The topological universality observed by Vazza & Feletti is a consequence of cost-minimizing network optimization under generic constraints (as we show in §11.4), which applies to Λ CDM structure formation

as well, but Λ CDM does not *derive* this universality from its own dynamics—it is an emergent property that the theory does not explain from first principles.

Section Result and Implications

The existing literature provides: (a) a documented empirical pattern (Vopson); (b) a mathematical framework for regularities vs. randomness (Gell-Mann–Lloyd); (c) a physical mechanism to be evaluated (gravitating vacuum); (d) unexplained observational features (cosmic web topology). This paper connects all four.

4 Definitional Framework

4.1 Kolmogorov complexity

Definition 4.1 (Kolmogorov complexity). For a string x (or a discretized field), the *Kolmogorov complexity* $K(x)$ is the length of the shortest program producing x on a universal Turing machine U :

$$K(x) = \min\{|p| : U(p) = x\}. \quad (1)$$

$K(x)$ is uncomputable in general [Kolmogorov, 1965, Chaitin, 1966] but can be approximated from above. The invariance theorem guarantees that $K(x)$ is independent of the choice of U up to an additive constant.

4.2 The Gell-Mann–Lloyd decomposition

Definition 4.2 (Effective complexity). The Kolmogorov complexity decomposes as

$$K(x) = K_{\text{eff}}(x) + K_{\text{rand}}(x), \quad (2)$$

where $K_{\text{eff}}(x)$ is the *effective complexity*—the length of a concise description of the regularities of x —and $K_{\text{rand}}(x)$ is the incompressible residual [Gell-Mann and Lloyd, 2003].

For a cosmological matter density field $\rho(\mathbf{x})$ discretized on N cells:

$$K_{\text{eff}}(\rho) = \min_{M \in \mathcal{M}} \{|M| : d(\rho, \rho_M) < \varepsilon\}, \quad (3)$$

where $\mathcal{M}$ is the class of admissible models, $|M|$ is description length, and d is a distance metric.

- Homogeneous field: $K_{\text{eff}} \approx 0$.
- Random field: $K_{\text{eff}} \approx 0$ (no regularities).
- Cosmic web: K_{eff} is *maximal*—many non-reducible regularities at many scales.

4.3 The definitional chain

Proposition 4.3 (Structure–information identity).

$$\textit{Structure} \equiv \textit{Compressible information} \equiv K_{\text{eff}}. \quad (4)$$

Proof. *Structure* = set of constraints on configurations. *Compressible information* = regularities encodable in a shorter program. K_{eff} = length of that program. The three refer to the same mathematical object. $\square$

Counterarguments and Responses

Objection: K_{eff} is uncomputable.

Response: K_{eff} can be approximated by the minimum description length (MDL) principle [Rissanen, 1978, Grünwald, 2007]. For cosmological fields, the number of parameters in the best-fit power spectrum model provides a practical estimate. The *relative change* ΔK_{eff} between states is well-defined.

Section Result and Implications

Measuring structure growth is equivalent to measuring effective complexity growth. The question is: what mechanism drives this growth, and does it converge?

5 The QCD Derivation of the Vacuum–Matter Coupling

This section derives the coupling α entirely from established nuclear physics. No cosmological data is used.

5.1 The chiral condensate and the GMOR relation

The QCD vacuum—the ground state of quantum chromodynamics, the Standard Model theory of the strong nuclear force—is characterized by a nonzero *chiral condensate* $\langle \bar{q}q \rangle \approx -(250 \text{ MeV})^3$ [Bazavov et al., 2012, Borsanyi et al., 2012]. This is the order parameter for spontaneous chiral symmetry breaking: the QCD vacuum “prefers” a state in which quark–antiquark pairs condense, lowering the energy.

The Gell-Mann–Oakes–Renner (GMOR) relation [Gell-Mann et al., 1968] connects the condensate to pion observables:

$$f_\pi^2 m_\pi^2 = -m_q \langle \bar{q}q \rangle, \quad (5)$$

where $f_\pi \approx 93 \text{ MeV}$ is the pion decay constant, $m_\pi \approx 140 \text{ MeV}$ is the pion mass, and $m_q = (m_u + m_d)/2 \approx 3.5 \text{ MeV}$ is the average light quark mass.

5.2 In-medium condensate at finite baryon density

At baryon number density n_B and zero temperature [Drukarev and Levin, 1991, Cohen et al., 1992]:

$$\frac{\langle \bar{q}q \rangle_{n_B}}{\langle \bar{q}q \rangle_0} = 1 - \frac{\sigma_\pi n_B}{f_\pi^2 m_\pi^2} + \mathcal{O}(n_B^2), \quad (6)$$

where σ_π is the *pion-nucleon sigma term*: the matrix element $\sigma_\pi = m_q \langle N | \bar{u}u + \bar{d}d | N \rangle$ measuring how much the nucleon's mass depends on the light quark masses. The linear term is leading order in chiral perturbation theory [Gasser et al., 1991]; corrections are suppressed by n_B/n_0 where $n_0 \approx 0.16 \text{ fm}^{-3}$ is nuclear saturation density. At all cosmological densities, $n_B/n_0 < 10^{-44}$, so the linear approximation is exact.

Experimental status of σ_π : $\sigma_\pi = 59.1 \pm 3.5 \text{ MeV}$ from pionic atom spectroscopy [Hoferichter et al., 2015]; $\sigma_\pi = 41\text{--}55 \text{ MeV}$ from lattice QCD [Yang et al., 2016, Alexandrou et al., 2020, Gupta et al., 2021]. We adopt $\sigma_\pi = 50 \pm 10 \text{ MeV}$.

The *strangeness sigma term* $\sigma_s = m_s \langle N | \bar{s}s | N \rangle \approx 40 \pm 10 \text{ MeV}$ [Junnarkar and Walker-Loud, 2013, Yang et al., 2016] measures the nucleon's coupling to the strange quark condensate. Strange quarks are *not* valence constituents of the nucleon (a proton is *uud*). The nonzero σ_s arises entirely from virtual $s\bar{s}$ fluctuations in the QCD vacuum sampled by the nucleon's gluon field. This is an unambiguous vacuum effect.

5.3 Equation of state of the condensate shift

Proposition 5.1 (Equation of state). *The chiral condensate $\langle \bar{q}q \rangle$ is a Lorentz scalar. Its vacuum expectation value contributes to the stress-energy tensor as*

$$T_{\text{cond}}^{\mu\nu} = -\rho_{\text{cond}} g^{\mu\nu}, \quad (7)$$

where $\rho_{\text{cond}} = -m_q \langle \bar{q}q \rangle$. This is the equation of state $p = -\rho$, i.e., $w = -1$.

Proof. In the QCD vacuum, Lorentz invariance requires $\langle T^{\mu\nu} \rangle_{\text{vac}} = -\rho_{\text{vac}} g^{\mu\nu}$ [Weinberg, 1989]. The chiral condensate is a Lorentz scalar: it has no preferred direction, no momentum flow, no anisotropic stress. A shift $\delta \langle \bar{q}q \rangle$ at finite density produces $\delta T_{\text{cond}}^{\mu\nu} = m_q \delta \langle \bar{q}q \rangle g^{\mu\nu}$. This is $w = -1$ by construction. $\square$

Remark 5.2. This distinguishes the condensate energy from the kinetic energy of quarks inside the nucleon, which has $T_{\text{kin}}^{\mu\nu} = (\rho + p) u^\mu u^\nu + p g^{\mu\nu}$ with $w > 0$. The sigma term quantifies the *scalar condensate* contribution, not the kinetic contribution.

Remark 5.3 (The in-medium condensate and the matter sector). The argument above is rigorous in vacuum but requires scrutiny at finite density. At finite baryon density $n_B > 0$, Lorentz invariance is broken by the presence of the medium (there is a preferred rest frame). The condensate shift $\delta \langle \bar{q}q \rangle$ is not a vacuum fluctuation—it is a response of the ground state to the presence of matter. Whether this response gravitates as $w = -1$ or is absorbed into the matter sector's effective equation of state is the central physical question of this paper.

In standard nuclear physics, the full nucleon mass m_N —including the sigma-term contribution—gravitates as $w = 0$ matter. This is the conventional treatment. Our

proposal reclassifies $\sim 10\%$ of m_N as $w = -1$. The justification is the Lorentz scalar structure of the condensate, but the objection that “it is already counted in m_N ” has force.

Recent lattice QCD work on the nucleon mass decomposition [Yang et al., 2018, Ji, 1995] separates m_N into quark mass terms ($\sigma_\pi + \sigma_s$), quark kinetic energy, gluon field energy, and the trace anomaly contribution. These components have distinct Lorentz structures. The quark mass terms, being proportional to $\langle \bar{q}q \rangle$, inherit the scalar structure of the condensate. Whether they can be *gravitationally* separated from the kinetic components is an open question that would require computing the gravitational form factors $A(t)$, $B(t)$, $D(t)$ of the nucleon with separate treatment of each mass component—a lattice calculation that has not been performed.

We therefore acknowledge that the equation-of-state assignment $w = -1$ for the condensate shift is the most vulnerable point of the derivation. It is physically motivated (Lorentz scalar structure), consistent with all observational data (the predicted α matches observations), and more principled than the alternative (ignoring the Lorentz structure and treating everything as $w = 0$). But it is not proved from first principles, and a definitive resolution requires lattice QCD input that does not yet exist.

5.4 The chiral vacuum energy shift

The explicit chiral symmetry breaking contribution to the vacuum energy:

$$\rho_{\text{vac}}^{(\text{chiral})} = -m_q \langle \bar{q}q \rangle = f_\pi^2 m_\pi^2. \quad (8)$$

At finite baryon density, substituting (6):

$$\Delta \rho_{\text{vac}}^{(\text{chiral})} = -m_q (\langle \bar{q}q \rangle_{n_B} - \langle \bar{q}q \rangle_0) = \frac{\sigma_\pi}{m_N} \rho_m, \quad (9)$$

where $\rho_m = m_N n_B$ is the baryonic mass density.

Including the strangeness contribution:

$$\alpha_{\text{bare}} = \frac{\sigma_\pi + \sigma_s}{m_N} \approx \frac{50 + 40}{938} \approx 0.096. \quad (10)$$

5.5 The trace anomaly consistency check

The QCD energy-momentum trace anomaly for $N_f = 3$ light flavors [Collins et al., 1977]:

$$\langle T^\mu{}_\mu \rangle = \frac{\beta(g)}{2g} \langle G_{\mu\nu} G^{\mu\nu} \rangle + \sum_f m_f \langle \bar{q}_f q_f \rangle. \quad (11)$$

In the nucleon state (Feynman–Hellmann theorem):

$$m_N = \sigma_\pi + \sigma_s + \sigma_{\text{heavy}} + E_{\text{gluon}}, \quad (12)$$

where $\sigma_{\text{heavy}} = \frac{2}{9}(m_N - \sigma_\pi - \sigma_s) \approx 188$ MeV [Shifman et al., 1978] and $E_{\text{gluon}} \approx 660$ MeV. At finite density:

$$\delta \langle T^\mu{}_\mu \rangle = m_N n_B = \rho_m. \quad (13)$$

The gluon condensate shift adds coherently to the quark shift—it does not cancel it. Both contributions increase $\langle T^\mu_\mu \rangle$ at finite density, as required by (13).

5.6 Physical reduction factors

Two physical effects reduce the bare coupling to the observable value:

1. Baryon fraction. The QCD condensate shift couples only to baryonic matter. If dark matter has negligible QCD interactions:

$$\alpha_{\text{eff}} = \alpha_{\text{bare}} \times \frac{\Omega_b}{\Omega_m} = 0.096 \times 0.156 = 0.015, \quad (14)$$

where $\Omega_b/\Omega_m = 0.156$ [Planck Collaboration, 2020].

2. Nonlinear self-screening. N-body simulations (§7) show that the density-dependent modification confines itself to the shrinking overdense volume fraction as structure grows. The screening factor at $z = 0$ is approximately 3 (resolution-convergent at 32^3 and 64^3):

$$\alpha_{\text{obs}} \approx \frac{\alpha_{\text{eff}}}{3} \approx 0.005. \quad (15)$$

5.7 Comparison with observations

Quantity	$ \alpha $	Source
QCD bare	0.096 ± 0.020	Eq. (10), $\sigma_\pi = 50 \pm 10$, $\sigma_s = 40 \pm 10$
× baryon fraction	0.015 ± 0.003	Planck Ω_b/Ω_m
÷ nonlinear screening	0.005 ± 0.003	N-body (§7), factor ~ 2 –4
Observed (best fit to $f\sigma_8$)	0.003	Structure growth data

Table 1: Predicted vs. observed vacuum–matter coupling. One parameter, derived from nuclear physics, with no fitting to cosmological data.

The predicted $|\alpha| = 0.005 \pm 0.003$ encompasses the observed value 0.003. The agreement is nontrivial—it connects nuclear physics measurements to cosmological structure growth—but we do not claim “zero free parameters” in the strong sense. The screening factor introduces an uncertainty of approximately a factor of 2.

Counterarguments and Responses

Objection: In standard nuclear physics, σ_π is included in m_N . When we write $\rho_m = m_N n_B$, this energy is already gravitating as $w = 0$ matter.

Response: The condensate contribution has $w = -1$ by its Lorentz structure (Proposition 5.1). Treating it as $w = 0$ is the approximation: the standard cosmological treatment neglects a $\sim 5\%$ $w = -1$ component. The correction is small enough to have been invisible in standard cosmology but large enough to be relevant for the S_8 tension at the $\sim 3\%$ level.

Objection: The factor-of-1.7 agreement is not compelling for a “prediction.”

Response: We agree this is not a precision test. It is an order-of-magnitude prediction connecting nuclear physics to cosmology via a chain with no adjustable parameters. The screening factor (the main source of uncertainty) can be refined with higher-resolution simulations (§16).

Section Result and Implications

The vacuum–matter coupling $\alpha \approx 0.005$ is derived from the in-medium chiral condensate shift, using experimentally measured sigma terms and the Planck baryon fraction. The gluon condensate adds coherently. The strangeness sigma term provides an unambiguous vacuum coupling. Agreement with the cosmological best fit is within the uncertainty band.

6 Modified Gravitational Dynamics

6.1 The vacuum–matter ansatz

If the cosmological constant Λ and the quantum vacuum energy ρ_{vac} are physically distinct—a separation supported by trace-free Einstein gravity, Kaloper–Padilla sequestering [Kaloper and Padilla, 2014], Volovik’s condensed-matter analogy [Volovik, 2003], and Solà Peracaula’s running vacuum [Solà, 2013]—then the vacuum energy responds to local matter density:

$$\rho_{\text{vac}}(\rho_m) = \Lambda_0 - \alpha \rho_m, \quad (16)$$

where Λ_0 is the unperturbed vacuum energy density and α is the coupling derived in §5.

6.2 Modified Poisson equation

The vacuum perturbation $\delta\rho_{\text{vac}} = -\alpha \delta\rho_m$ with $w = -1$ contributes to the gravitational source as $\delta\rho_{\text{vac}} + 3\delta p_{\text{vac}}/c^2 = -2\delta\rho_{\text{vac}} = 2\alpha \delta\rho_m$:

$$\nabla^2\Phi = 4\pi G(1 + 2\alpha) \rho_m. \quad (17)$$

The background Friedmann equation is *identical* to Λ CDM:

$$H^2 = \frac{8\pi G}{3} \bar{\rho}_m + \frac{\Lambda_{\text{geom}}}{3}. \quad (18)$$

CMB peak positions, BAO, and $H(z)$ are unaffected. The modification enters only through the Poisson equation for perturbations in overdense regions.

6.3 Two-phase dynamics

Expanding phase (voids): $\rho_m \rightarrow 0$, $\rho_{\text{vac}} \approx \Lambda_0$. Source: $\rho_{\text{vac}} + 3p/c^2 = -2\rho_{\text{vac}} < 0$. Voids actively repel matter.

Bound phase (clusters, filaments): $\rho_m \gg 0$, vacuum suppressed, $G_{\text{eff}} = G(1 + 2\alpha)$. Vacuum gravitates attractively. Transition at:

$$r_{\text{vac}} = \left(\frac{2GM}{H_v^2} \right)^{1/3}, \quad H_v = \sqrt{\frac{8\pi G \rho_{\text{vac}}}{3}}. \quad (19)$$

Section Result and Implications

The gravitating vacuum creates an asymmetric two-phase medium: repulsive voids that actively expel matter, and attractive bound structures with enhanced gravity. This goes beyond passive gravitational instability.

7 N-body Simulations and the Discovery of Nonlinear Self-Screening

7.1 The particle-mesh algorithm

We solve the modified N-body problem using a particle-mesh (PM) algorithm [Hockney and Eastwood, 1981]. The only modification to the standard PM scheme is algebraic: in overdense cells ($\delta_i > 0$), the density contrast is multiplied by $(1 + 2\alpha)$ before solving the Poisson equation. In underdense cells ($\delta_i \leq 0$), standard gravity applies:

$$G_{\text{eff},i} = \begin{cases} G(1 + 2\alpha) & \text{if } \delta_i > 0, \\ G & \text{if } \delta_i \leq 0. \end{cases} \quad (20)$$

The background cosmology is identical to Λ CDM. The modification is a single algebraic change to the Poisson source.

7.2 Initial conditions and parameters

- Box size: $L = 200 h^{-1}$ Mpc (comoving).
- Grid: 32^3 and 64^3 (convergence study).
- Cosmological parameters: $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km/s/Mpc, $\sigma_8 = 0.811$, $n_s = 0.965$ [Planck Collaboration, 2020].
- Initial redshift: $z_i = 50$.
- Values of α tested: 0 , ± 0.003 , ± 0.01 , ± 0.03 .

Initial conditions are generated from the Zel'dovich approximation with the Λ CDM transfer function.

7.3 Convergence study

The σ_8 ratio ($\alpha \neq 0$ vs. $\alpha = 0$) at $z = 0$ changes by less than 1.5% between 32^3 and 64^3 resolutions for $|\alpha| \leq 0.03$. The power spectrum enhancement is converged to $< 3\%$ at $k < 0.5 h/\text{Mpc}$. The density-threshold sensitivity is also small: varying the threshold from $\delta = -0.5$ to $\delta = +0.5$ changes σ_8 by $< 1.5\%$.

7.4 The discovery of nonlinear self-screening

The central finding: the σ_8 enhancement at $z = 0$ is approximately **three times smaller** than linear perturbation theory predicts.

Linear theory gives $G_{\text{eff}} = G(1 + 2\alpha)$ everywhere, predicting $\sigma_8^{(\alpha)}/\sigma_8^{(0)} = (1 + 2\alpha)^{0.55}$ [Linder, 2005]. For $\alpha = 0.03$: linear prediction = 1.033; simulation result = 1.011. The ratio: $0.011/0.033 = 0.33 \approx 1/3$.

Physical origin: At early times ($z > 5$), the modification applies nearly uniformly (most cells are near $\delta \approx 0$). As structure grows, overdense cells become a smaller fraction of the total volume. By $z = 0$, the modification is locked inside halos and filaments occupying $\sim 20\text{--}30\%$ of the volume. The volume-weighted effect is suppressed by an estimated factor of ~ 3 , though this estimate carries substantial uncertainty ($\sim 2\text{--}5$) due to the limited resolution of the present simulations.

This is not a fine-tuning. It is a natural consequence of the density-dependent threshold. However, the precise numerical value of the screening factor is *not* established by these simulations: convergence between 32^3 and 64^3 is necessary but not sufficient for convergence at 512^3 or higher, particularly for a nonlinear effect sensitive to the volume fraction of overdense cells. We treat the factor of ~ 3 as an order-of-magnitude estimate, not a precision measurement.

Counterarguments and Responses

Objection: 32^3 and 64^3 are modest resolutions. The screening factor may change at higher resolution.

Response: We agree that higher-resolution simulations ($\geq 512^3$) are a priority (§16). The convergence between 32^3 and 64^3 is encouraging but not definitive. However, the physical mechanism (volume confinement of the overdense fraction) is generic and does not depend on resolution—only the precise numerical value of the screening factor does.

Section Result and Implications

N-body simulations reveal nonlinear self-screening: the σ_8 enhancement is $\sim 3\times$ smaller than linear theory predicts. The screening is a natural consequence of structure growth, not a fine-tuning. It reduces $\alpha_{\text{eff}} = 0.015$ to $\alpha_{\text{obs}} \approx 0.005$, in agreement with the cosmological best fit.

8 The Two Information Bits

8.1 The infodynamic bit (Vopson)

Definition 8.1 (Infodynamic bit). The mass of one bit of information at thermal equilibrium [Vopson, 2019, Landauer, 1961]:

$$m_{\text{bit}}^{\text{TD}} = \frac{k_B T \ln 2}{c^2}. \quad (21)$$

At $T = 300 \text{ K}$: $m_{\text{bit}}^{\text{TD}} \approx 3.19 \times 10^{-38} \text{ kg}$. This is a *universal lower bound* on the mass-energy cost of information processing.

8.2 The QCD bit

Definition 8.2 (QCD bit). For a baryonic system of mass M_{baryon} with effective complexity $K_{\text{eff}}(\rho)$:

$$M_{\text{grav}} = (1 + 2\alpha) M_{\text{baryon}}, \quad (22)$$

$$\Delta M_{\text{structure}} = 2\alpha M_{\text{baryon}}, \quad (23)$$

$$m_{\text{bit}}^{\text{QCD}} = \frac{2\alpha M_{\text{baryon}}}{K_{\text{eff}}(\rho)}. \quad (24)$$

The QCD bit is *system-specific*: it gives the gravitational mass contributed by each bit of structural complexity in a bound system, derived from the chiral condensate shift.

8.3 How they relate

The infodynamic bit is **universal** (depends only on T): minimum mass-energy cost of one bit. The QCD bit is **system-specific** (depends on M_{baryon} and K_{eff}): actual gravitational contribution of one bit of structure.

Together: Vopson’s bit sets the thermodynamic floor; the QCD bit gives the gravitational ceiling. For a galaxy with $M_{\text{baryon}} = 10^{12} M_{\odot}$ and $K_{\text{eff}} \sim 10^{20}$ bits: $m_{\text{bit}}^{\text{QCD}} \sim 10^{-10} M_{\odot}$ —vastly larger than the thermodynamic floor.

Counterarguments and Responses

Objection: These are different quantities with different origins. Calling both “mass of a bit” is misleading.

Response: They are explicitly labeled as different: the infodynamic bit is the cost of *processing* one bit; the QCD bit is the gravitational *consequence* of one bit of structure. They are complementary.

Section Result and Implications

Information has gravitational mass in two complementary senses: a universal thermodynamic minimum (Vopson) and a system-specific QCD contribution (this work). The latter is derived from measured quantities.

9 Simulability as an Epistemological Criterion

9.1 Not a bug—a feature

Vopson [Vopson, 2023] proposed that the universe behaves like a computational system. Critics treated this as unfalsifiable metaphysics. We argue that the computational property, properly defined, is a *measurable, falsifiable feature* of physical theories.

Falsifiability (Popper, 1959) asks: can this theory be proved wrong? But falsifiability says nothing about the *informational structure* of a theory.

We propose that **simulability** fills this gap. Simulability is not binary—it is a *spectrum*. A theory is *more simulable* the less its predictions depend on initial conditions.

Definition 9.1 (Simulability). A physical system with dynamics Φ on a complete metric space S has simulability degree

$$\mathcal{S} = 1 - k, \quad (25)$$

where k is the Lipschitz constant of Φ . If $k < 1$: Φ has a unique fixed point; initial conditions are forgotten at rate k^n .

- $\mathcal{S} = 1$ ($k = 0$): maximally simulable. One-step convergence.
- $\mathcal{S} = 0$ ($k = 1$): not simulable. Outcome entirely from initial conditions.
- $0 < \mathcal{S} < 1$: partially simulable. Information from initial conditions decays as k^n .

9.2 Application

The gravitating vacuum has $k \ll 1$ (§10), giving $\mathcal{S} \approx 0.99$ *conditional on Condition C*. If Condition C holds, the cosmic web’s topology is determined almost entirely by the laws (α, G, Λ_0) . The random seed determines the path to the fixed point, not the fixed point itself.

Λ CDM predicts robust *statistical* properties (power spectrum, halo mass function) with $\mathcal{S}_{\text{stat}} \approx 1$ for ensemble-level quantities. For individual configurations (specific filament locations), $\mathcal{S}_{\text{ind}} \approx 0$: these depend on the random seed. The gravitating vacuum, if Condition C holds, would have high simulability for *both* levels—but this comparison is between an established result (Λ CDM ensemble predictions) and a conditional one (individual predictions contingent on unproven mathematics). The comparison becomes meaningful only when Condition C is either proved or refuted.

9.3 What simulability reveals

The informational nature of physics. A simulable system’s laws contain enough information to determine the outcome.

Why information entropy decreases. The contraction mapping converges to a maximally structured fixed point. “Data compression” is the inevitable consequence.

Falsifiability. Simulability predicts suppressed cosmic variance and deterministic filament continuation (§12).

A hierarchy of theories. Λ CDM is falsifiable but has low individual simulability. The gravitating vacuum is both falsifiable and highly simulable. The informational content of the laws *replaces* the informational content of initial conditions.

Section Result and Implications

Simulability $\mathcal{S} = 1 - k$ is a measurable property of physical theories. It quantifies the degree to which laws determine outcomes independently of initial conditions. This is the rigorous content of Vopson’s “simulation” property.

10 Variational Principle and Contraction Mapping

10.1 The cosmic web functional

The cosmic web minimizes:

$$E[\rho_m] = \int_V \rho_{\text{vac}} dV + \oint_{\partial V} \sigma_{\text{eff}} dA + \int_B \Phi_{\text{grav}} dV, \quad (26)$$

with $\sigma_{\text{eff}} \propto \alpha \rho_m^{\text{wall}} \cdot d$ from the Israel junction conditions [Israel, 1966]. Minimizers satisfy generalized Plateau’s laws: walls meet in threes at 120° ; filaments in fours at $\sim 109.5^\circ$.

10.2 The cyclical operator

$$\Phi_{\text{cycle}} = F_3 \circ F_2 \circ F_1, \quad (27)$$

$$F_1 : \rho_m \mapsto \rho_{\text{vac}}(\rho_m) \quad (\text{vacuum assignment}), \quad (28)$$

$$F_2 : \rho_{\text{vac}} \mapsto g_{\mu\nu} \quad (\text{Poisson/Einstein solver}), \quad (29)$$

$$F_3 : g_{\mu\nu} \mapsto \rho'_m \quad (\text{matter redistribution}). \quad (30)$$

10.3 Lipschitz estimates for each operator

F_1 : Linear map with $\|F_1(\rho_1) - F_1(\rho_2)\| = \alpha \|\rho_1 - \rho_2\|$. Lipschitz constant: $L_1 = \alpha$.

F_2 : The Poisson equation $\nabla^2 \Phi = 4\pi G \rho$ is linear; the Green’s function operator has bounded norm $\|G_P\| \sim C_\Phi$ where C_Φ is of order unity (depends on Ω_m , Ω_Λ , and the box size). In the linear regime, $L_2 = C_\Phi \sim \mathcal{O}(1)$.

Nonlinear caveat: In the full nonlinear problem, collapsed regions develop arbitrarily large density contrasts. The Lipschitz constant of F_2 on the full nonlinear space $L^2(\Omega)$ is not bounded. However, on the subspace of density fields with bounded density contrast $|\delta| < \delta_{\text{max}}$ (which is the physically relevant regime at any finite time), L_2 remains bounded.

Remark 10.1 (Completeness of the restricted space). The subspace $S_{\delta_{\max}} = \{\rho \in L^2(\Omega) : |\delta(\mathbf{x})| < \delta_{\max}\}$ is *not closed* in $L^2(\Omega)$: a Cauchy sequence of fields with bounded contrast can converge to a field with unbounded contrast (e.g., approaching a delta function). Strictly, the Banach contraction mapping theorem requires a complete metric space, which $S_{\delta_{\max}}$ is not.

Two resolutions are possible: (i) work on the *closure* $\bar{S}_{\delta_{\max}}$, which is complete, and show that Φ_{cycle} maps $\bar{S}_{\delta_{\max}}$ into itself (requiring that the dynamics do not produce singularities in finite time—plausible for $\alpha \ll 1$ but unproven); or (ii) use a weighted Sobolev space that naturally penalizes large density contrasts, making the relevant subspace complete. Both approaches require functional-analytic work beyond the scope of this paper. We identify this as an open problem alongside the Lipschitz bound itself.

For the purposes of this paper, Condition C should be understood as including the implicit assumption that the Picard iterates remain in a space where the contraction bound holds. This is consistent with all numerical evidence but is not proved analytically.

F_3 : Matter redistribution along geodesics involves dissipation through violent relaxation [Binney and Tremaine, 2008]. The Lipschitz constant is $L_3 = (1 - \varepsilon_{\text{grav}})$ where $\varepsilon_{\text{grav}} \in (0.1, 0.5)$.

10.4 Composite contraction factor

$$k = L_3 \cdot L_2 \cdot L_1 = (1 - \varepsilon_{\text{grav}}) \cdot C_{\Phi} \cdot \alpha. \quad (31)$$

For $\alpha = 0.005$, $C_{\Phi} \sim 1$, $\varepsilon_{\text{grav}} \sim 0.3$:

$$k \approx 0.7 \times 1 \times 0.005 = 0.0035 \ll 1. \quad (32)$$

The contraction is driven primarily by $\alpha \ll 1$; dissipation provides additional margin.

10.5 The contraction condition

We state the contraction as a condition rather than an unconditional theorem:

Condition 10.2 (Contraction Condition C). *The operator Φ_{cycle} satisfies $\|\Phi_{\text{cycle}}(\rho_1) - \Phi_{\text{cycle}}(\rho_2)\|_{L^2} \leq k \|\rho_1 - \rho_2\|_{L^2}$ with $k < 1$ on the space of density fields with bounded density contrast.*

Status of Condition C:

- *Proved* in the linear regime ($|\delta| \ll 1$): the Lipschitz estimates above are rigorous.
- *Verified numerically, indirectly* in the nonlinear regime: N-body simulations (§7) show convergent power spectra for $|\alpha| \leq 0.03$ across two resolutions. However, this verification is indirect: it demonstrates stability and convergence of statistical measures, not convergence of different initial conditions to the same fixed point. A direct test—running identical boxes with different random seeds and measuring convergence of topological statistics (filling factor, node degree, clustering coefficient)—has not been performed and is identified as a priority in §16.

- *Not proved* for the full unbounded nonlinear space: shell-crossing, caustic formation, and multi-streaming are not Lipschitz continuous in L^2 . The restricted space $S_{\delta_{\max}}$ is not complete (Remark 10.1).

Theorem 10.3 (Existence and uniqueness of the fixed point). *Under Condition C, there exists a unique density field ρ^* such that $\Phi_{\text{cycle}}(\rho^*) = \rho^*$. For any initial field ρ_0 :*

$$\|\Phi^n(\rho_0) - \rho^*\| \leq k^n \|\rho_0 - \rho^*\|. \quad (33)$$

Proof. By the Banach contraction mapping theorem [Banach, 1922]: on a complete metric space, a contraction has a unique fixed point with geometric convergence. Condition C provides the contraction. $L^2(\Omega)$ is complete. $\square$

The fixed point ρ^* is the cosmic web. Its topological properties are determined by α alone.

Counterarguments and Responses

Objection: Presenting Theorem 10.3 as a theorem when Condition C is unproven in the full nonlinear case is misleading.

Response: We explicitly state Condition C as a condition, not an assumption. The theorem is conditional: “under Condition C.” This is standard mathematical practice—many important results in physics are conditional on hypotheses verified numerically but not proved analytically (e.g., the assumption of ergodicity in statistical mechanics). The paper clearly distinguishes between what is proved (the linear case) and what is numerically verified (the nonlinear case).

Section Result and Implications

Under Condition C (proved linearly, verified numerically), the cosmic web is the unique fixed point of the matter–vacuum–metric dynamics. The contraction factor $k \approx 0.004 \ll 1$ implies near-maximal simulability $\mathcal{S} \approx 0.996$.

11 Proof of the Second Law of Infodynamics

11.1 The monotonicity lemma

Lemma 11.1 (Monotonicity of K_{eff} along the energy-decreasing path). *Let $\rho_n = \Phi^n(\rho_0)$ be the Picard iterates. If the variational energy $E[\rho]$ (Eq. 26) satisfies $E[\rho_{n+1}] < E[\rho_n]$, then $K_{\text{eff}}(\rho_{n+1}) > K_{\text{eff}}(\rho_n)$.*

Proof. Reducing $E[\rho]$ requires redistributing matter from a more homogeneous to a less homogeneous configuration:

1. The volume term $\int_V \rho_{\text{vac}} dV$ is minimized by emptying voids (creating underdensities).

2. The binding term $\int_B \Phi_{\text{grav}} dV$ is minimized by concentrating clusters (creating overdensities).

3. The surface term $\oint \sigma_{\text{eff}} dA$ penalizes irregular boundaries (favoring smooth walls).

Each creates density contrasts absent in ρ_n : new walls, sharper filaments, deeper voids. These are new regularities—encodable in a model M with additional parameters. The fit residual $d(\rho_{n+1}, \rho_M) < \varepsilon$ requires capturing the new contrasts, so the model description length $|M|$ for ρ_{n+1} exceeds that for ρ_n . By definition (3), $K_{\text{eff}}(\rho_{n+1}) > K_{\text{eff}}(\rho_n)$.

The only energy-decreasing path that would *not* increase K_{eff} is uniform contraction (all densities scale equally). This is forbidden by conservation of total mass: $\int_{\Omega} \rho_m dV = \text{const}$. Uniform scaling violates this unless the scaling factor is unity. $\square$

Remark 11.2 (Status of the proof). This proof is physical rather than formally complete. Two specific gaps remain:

(a) Parameter sharpening vs. new parameters. The argument assumes that energy decrease creates *new* regularities requiring additional model parameters. But a scenario exists where existing structures sharpen (filaments become thinner, voids become emptier) without creating topologically new features. In that case, the same model with adjusted parameter *values* suffices, and K_{eff} (which counts the number of parameters, not their precision) could remain constant while $E[\rho]$ decreases. The lemma therefore holds strictly only for iterations that create *new topological features* (new filaments, new voids, new nodes)—not for iterations that merely refine existing ones. We conjecture, but do not prove, that the majority of iterations along the Picard sequence belong to the former category during the structure-formation epoch.

(b) Formal measure-theoretic link. A rigorous proof would establish that K_{eff} is a monotone function of the L^2 distance to the fixed point along the specific trajectory of Φ_{cycle} . This requires functional-analytic methods beyond the scope of this paper and is identified as a priority for future work (§16).

The present argument is physically motivated and consistent with all numerical evidence from the N-body simulations, but it is not a complete proof. Theorem 11.3 inherits this limitation: it is conditional on Lemma 11.1 in addition to Condition C.

11.2 The Second Law

Theorem 11.3 (Second Law of Infodynamics—conditional form). *Under Condition C and Lemma 11.1, the effective complexity of the matter density field increases monotonically:*

$$K_{\text{eff}}(\Phi^{n+1}(\rho_0)) \geq K_{\text{eff}}(\Phi^n(\rho_0)), \quad (34)$$

converging to $K_{\text{max}}^ = K_{\text{eff}}(\rho^*)$. Equality holds only at the fixed point.*

Proof. By Condition C, $\|\rho_{n+1} - \rho^*\| < \|\rho_n - \rho^*\|$. The contraction brings ρ_n monotonically toward ρ^* . The variational principle (26) guarantees $E[\rho_{n+1}] < E[\rho_n]$ (the fixed point is the energy minimum). By Lemma 11.1, $K_{\text{eff}}(\rho_{n+1}) > K_{\text{eff}}(\rho_n)$. Convergence follows from continuity of K_{eff} and uniqueness of ρ^* . $\square$

In words: *under the stated conditions, the structural complexity of gravitating systems increases monotonically until the unique equilibrium is reached.*

11.3 Corollary: Shannon entropy decreases

Remark 11.4 (Shannon entropy decreases—approximate). The coarse-grained Shannon entropy satisfies approximately:

$$H_{\text{coarse}}(\rho_n) \approx H_{\text{total}} - K_{\text{eff}}(\rho_n). \quad (35)$$

Since K_{eff} increases (Theorem 11.3) and H_{total} is approximately constant, H_{coarse} decreases. This recovers Vopson’s empirical observation as an approximate consequence of the conditional theorem. The approximation (35) relates Shannon entropy and Kolmogorov complexity, which are asymptotically equivalent for ergodic stationary processes but not strictly valid for the cosmological density field (which is neither stationary nor ergodic in the relevant sense). This observation should therefore be regarded as physically motivated rather than rigorously derived.

11.4 Topological universality

Theorem 11.5 (Universality). *Any three-dimensional network minimizing*

$$F[G] = \sum_{\text{nodes}} C_{\text{node}} + \sum_{\text{edges}} C_{\text{edge}} \cdot L_{\text{edge}} + \lambda(\text{connectivity constraint}) \quad (36)$$

converges to the same topological class: degree-3 nodes, 120° angles, clustering coefficient ~ 0.4 –0.6.

This follows from the universality of the Steiner tree problem in three dimensions [Mézard and Montanari, 2009]. The class includes: the cosmic web ($C_{\text{node}} = E_{\text{grav}}$, $C_{\text{edge}} = \sigma_{\text{eff}}$); neuronal networks ($C_{\text{node}} = E_{\text{metabolic}}$, $C_{\text{edge}} = \gamma_{\text{axon}}$); slime mold ($C_{\text{edge}} \propto$ Poiseuille resistance) [Tero et al., 2010, Burchett et al., 2020].

Remark 11.6. This theorem applies equally to Λ CDM structure formation—the cost-minimizing universality is not specific to the gravitating vacuum. What the gravitating vacuum adds is the *uniqueness* of the fixed point (Theorem 10.3): not just that the topology is universal, but that the specific configuration is unique and independent of initial conditions.

Counterarguments and Responses

Objection: If topological universality applies to Λ CDM as well, this is not evidence for the gravitating vacuum.

Response: Correct. Topological universality (Theorem 11.5) is a mathematical property of cost-minimizing networks, not a discriminating test between models. The discriminating tests are Predictions 7 and 8 in §12: suppressed cosmic variance and supra-horizon individual coherence, which follow from the *uniqueness* of the fixed point, not from the topology alone.

Section Result and Implications

The Second Law of Infodynamics is derived as a conditional theorem: under Condition C (proved linearly, verified numerically) and Lemma 11.1 (physically rigorous), K_{eff} increases monotonically. The decrease of Shannon entropy follows as a corollary. The paper clearly distinguishes between the conditional theorem and an unconditional proof.

12 Falsifiable Predictions

Eight predictions, distinguishable from Λ CDM:

1. **Monotonic $K_{\text{eff}}(z)$.** Effective complexity grows with cosmic time, estimable from $P(k)$ as $K_{\text{eff}} \propto N_{\text{param}}(P_{\text{reg}}) \times \log(S/N)$.
2. **Web topology independent of initial conditions.** Compare web statistics across independent survey sub-volumes. Λ CDM predicts scatter from cosmic variance; this model predicts identical statistics.
3. **Deterministic filament continuation.** Predict structure in newly surveyed volumes from adjacent volumes, with accuracy exceeding Λ CDM cosmic variance.
4. **Excess of large voids.** Void size function shows excess of large voids relative to Press–Schechter. Testable with DESI/Euclid.
5. **Enhanced peculiar velocities near void boundaries.** Vacuum repulsion produces higher tangential velocities near voids. Observable in DESI redshift-space distortions.
6. **Web topology at $z > 6$.** Fully formed (compressed scale) at $z > 6$, testable with JWST + SKA 21 cm at $z = 10$ –20. Λ CDM predicts progressive formation.
7. **Suppressed cosmic variance.** Web statistics show less scatter across independent volumes than Λ CDM realizations. Testable by comparing ≥ 10 DESI/Euclid sub-volumes.
8. **Supra-horizon individual coherence.** Specific filaments continuous across causally disconnected volumes. Testable by tracking filaments across DESI–Euclid boundaries at $z \sim 2$ –4.

Predictions 7 and 8 are “discriminating” tests: they follow directly from Theorem 10.3 and have no natural explanation in Λ CDM (which predicts individual configurations determined by the random seed).

Section Result and Implications

Eight falsifiable predictions distinguish this framework from Λ CDM. The most decisive (7 and 8) test the uniqueness of the fixed point, not just the modified gravity.

13 Empirical and Data-Grounded Illustrations

13.1 The gravity–information equivalence

Equation (17) reads in two directions:

$$\nabla^2\Phi = 4\pi G(1 + 2\alpha)\rho_m \iff \text{gravitational potential maximizes } K_{\text{eff}}. \quad (37)$$

Gravitational clustering *is* the process that increases effective complexity.

13.2 The mass of structure

For the Milky Way: $M_{\text{baryon}} \approx 6 \times 10^{10} M_{\odot}$ [Bland-Hawthorn and Gerhard, 2016], $\alpha = 0.005$:

$$\Delta M_{\text{structure}} = 2 \times 0.005 \times 6 \times 10^{10} = 6 \times 10^8 M_{\odot}. \quad (38)$$

This is a small fraction of the total “dark mass” ($\sim 10^{12} M_{\odot}$). The QCD bit captures the microscopic condensate contribution. The bulk of the missing mass in the gravitating vacuum framework arises from the macroscopic vacuum capture mechanism at Level 2 [Kriger, 2026f]—a distinct effect not derived in this paper.

13.3 Void statistics

Characteristic void radius: $R_{\text{char}} \approx 22 h^{-1} \text{ Mpc}$ for $\alpha = 0.005$, consistent with SDSS ($17 h^{-1} \text{ Mpc}$ median) [Pan et al., 2012] and BOSS (20–100 $h^{-1} \text{ Mpc}$) [Mao et al., 2017].

14 Discussion

14.1 What is proved, what is derived, what is conjectured

14.2 On the falsifiability–simulability axis

Physical theories can be assessed along two independent axes:

- **Falsifiability:** Can the theory be proved wrong? (Popper)
- **Simulability:** How much of the outcome is determined by the laws? (This work)

Λ CDM: falsifiable, low individual simulability. The gravitating vacuum: falsifiable *and* highly simulable. A theory with higher simulability requires less input—it is a deeper theory.

Claim	Status	Section
$\alpha_{\text{bare}} = (\sigma_\pi + \sigma_s)/m_N$	Derived from QCD	§5
Condensate shift has $w = -1$	Lorentz argument (open: in-medium)	§5.3, Rem. 5.3
$\alpha_{\text{obs}} = 0.005 \pm 0.003$	Derived + numerical	§5, §7
Condition C (linear regime)	Proved	§10
Condition C (nonlinear)	Numerically verified (indirect)	§7
Unique fixed point ρ^*	Theorem (conditional on C)	§10
Lemma 11.1	Physical argument (gap: sharpening)	§11
Second Law (Thm. 11.3)	Conditional on C + Lemma	§11
Shannon entropy decreases	Approximate observation	§11, Rem. 11.4
Topological universality	Theorem	§11.4
$\mathcal{S} \approx 0.99$	Conditional on C	§9

Table 2: Status of each claim.

15 Limitations

1. **Condition C in the nonlinear regime.** Not proved analytically. Shell-crossing and caustic formation are not Lipschitz continuous. The restricted space $\mathcal{S}_{\delta_{\text{max}}}$ is not complete in L^2 (Remark 10.1).
2. **Lemma 11.1: parameter sharpening.** Energy decrease via sharpening existing structures (without creating new topological features) need not increase the model parameter count, and hence need not increase K_{eff} (Remark 11.2a).
3. **Lemma 11.1: formal status.** Physical argument, not measure-theoretic proof (Remark 11.2b).
4. **Screening factor.** Estimated as ~ 3 from 64^3 simulations with uncertainty ~ 2 –5; convergence at $\geq 512^3$ is not established.
5. $(1+z)^3$ **scaling of Λ_0 .** Motivated, not derived from a covariant action.
6. **Separation of Λ and ρ_{vac} .** Supported by five frameworks, not directly tested.
7. **Eq. (35).** The Shannon–Kolmogorov relationship is approximate and requires stationarity/ergodicity assumptions not strictly satisfied by the cosmological density field.
8. **Simulability comparison.** The claim $\mathcal{S} \approx 0.99$ for the gravitating vacuum is conditional on Condition C; the comparison with ΛCDM ($\mathcal{S}_{\text{ind}} \approx 0$) compares a conditional result with an established one.

16 Future Directions

1. Prove Condition C analytically, including the completeness issue (Remark 10.1): either on the closure $\bar{S}_{\delta_{\max}}$ or in a weighted Sobolev space.
2. Strengthen Lemma 11.1: prove that the parameter-sharpening mode (Remark 11.2a) is measure-zero along the Picard trajectory, or reformulate K_{eff} to count parameter precision as well as parameter number.
3. Run $\geq 512^3$ N-body simulations to establish the screening factor with $< 30\%$ uncertainty.
4. **Direct numerical test of Condition C:** Run identical simulation boxes with different random seeds and measure convergence of topological statistics (filling factor, node degree, clustering coefficient) to test whether different initial conditions produce the same fixed point—a direct test of the contraction mapping, as opposed to the indirect evidence from power spectrum convergence.
5. Compute $K_{\text{eff}}(z)$ from existing power spectrum data (SDSS, BOSS) as a first observational test.
6. **Lattice QCD gravitational form factors:** Compute the gravitational form factors $A(t)$, $B(t)$, $D(t)$ of the nucleon with separate treatment of the scalar condensate component, to determine whether the $w = -1$ assignment of Proposition 5.1 can be confirmed or refuted from first principles (see Remark 5.3).
7. Apply simulability to other physical theories and explore philosophical implications.
8. Investigate connection to Friston’s free energy principle [Friston, 2010].

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A Peer Reviews and Revision History

First Round

Reviewer recommendation: Major revision.

Major Issue 1 (Core proof has logical gaps): The reviewer noted that Lemma 11.1 is the load-bearing step and remains a physical argument, not a formal proof. The claim that every energy-decreasing step increases K_{eff} is asserted rather than demonstrated. A uniform contraction of the density field would decrease energy without adding new regularities; the paper’s dismissal via mass conservation is insufficient.

Response: Accepted. The revised version (a) explicitly acknowledges the gap; (b) presents Theorem 11.3 as a conditional theorem; (c) identifies the measure-theoretic formalization as an open problem.

Major Issue 2 (Banach contraction not established for full nonlinear system): F_2 is nonlinear with unbounded Lipschitz constant in collapsed regions; F_3 involves shell-crossing and multi-streaming, which are not Lipschitz in L^2 .

Response: Accepted. The contraction property is now stated as Condition C (10.2), proved in the linear regime and numerically verified in the nonlinear regime. The paper explicitly distinguishes between these two levels.

Major Issue 3 (Self-referential dependence on unpublished monograph): Key intermediate results existed only in the author’s monograph.

Response: The revised paper includes the full QCD derivation (§5), N-body methodology and results (§7), and contraction mapping analysis (§10) as self-contained sections.

Major Issue 4 (α derivation conflates steps of different rigor): The screening factor of ~ 3 from modest simulations introduces substantial uncertainty. “Zero free parameters” is an overstatement.

Response: Accepted. The revised version includes an uncertainty table (Table 1) with $\alpha_{\text{predicted}} = 0.005 \pm 0.003$. The phrase “zero free parameters” is replaced by “one parameter, derived from nuclear physics, with no fitting to cosmological data.”

Major Issue 5 (Simulability criterion is circular): $\mathcal{S} = 1 - k$ uses the unproven contraction constant. Characterizing Λ CDM as $\mathcal{S} = 0$ is misleading— Λ CDM predicts robust statistical properties.

Response: Partially accepted. The revised version distinguishes $\mathcal{S}_{\text{stat}}$ (ensemble) from $\mathcal{S}_{\text{ind}}$ (individual) and notes that Λ CDM has $\mathcal{S}_{\text{stat}} > 0$.

Major Issue 6 (Λ CDM explanatory failures overstated): Supra-horizon coherence arises from inflationary power spectrum; topological universality applies equally to Λ CDM; JWST tension has conventional explanations under investigation.

Response: Accepted. The abstract and §3 now state “explains statistically but not individually.” Remark 11.7 concedes that topological universality applies to Λ CDM as well.

Minor Issues: Recommended Journal section; weak counterarguments; Eq. (35) conflation; ΔM framing; preprint identification.

Response: Recommended Journal section retained in first revision (later removed). Eq. (35) given appropriate caveats. ΔM estimate contextualized with Level 2 vacuum capture.

Second Round

Reviewer recommendation: Minor-to-moderate revision.

Acknowledged improvements: Honest epistemic labeling (Table 2); Condition C; self-containment; fair treatment of Λ CDM; Lemma flagging.

Remaining Issue 1 (Parameter-sharpening gap): Existing filaments sharpening without new topological features could decrease $E[\rho]$ without increasing K_{eff} . The lemma conflates “new density contrasts” with “new regularities requiring additional model parameters.”

Response: Accepted. Remark 11.2(a) now explicitly names this gap and conjectures (without proof) that topology-creating iterations dominate during structure formation.

Remaining Issue 2 (Completeness of restricted space): $S_{\delta_{\text{max}}}$ is not closed in $L^2(\Omega)$; Banach theorem requires a complete metric space.

Response: Accepted. New Remark 10.1 addresses the issue, identifies two possible resolutions (closure with invariance, or weighted Sobolev spaces), and folds the completeness assumption into Condition C.

Remaining Issue 3 (N-body simulations underpowered): 32^3 – 64^3 is modest; convergence at these resolutions does not guarantee convergence at 512^3 .

Response: Partially accepted. Screening factor uncertainty widened to ~ 2 – 5 . The factor is now described as “an order-of-magnitude estimate, not a precision measurement.” Higher-resolution simulations identified as priority in §16.

Remaining Issue 4 (Simulability comparison): Comparing conditional $\mathcal{S} \approx 0.99$ with established Λ CDM predictions is comparing different things.

Response: Accepted. Section 9.2 rewritten to state explicitly that the comparison “becomes meaningful only when Condition C is either proved or refuted.”

Minor Issues: Remove Recommended Journal section; replace “smoking-gun.”

Response: Section removed. “Smoking-gun” replaced by “discriminating.”

Third Round

Reviewer recommendation: Accept with minor revisions.

Acknowledged improvements: Parameter-sharpening gap openly stated; completeness issue acknowledged; simulability comparison properly qualified; screening fac-

tor uncertainty widened; journal section removed; language softened. Table 2 praised as “a model of epistemic transparency.”

Remaining Issue 1 (In-medium condensate equation of state): Proposition 5.1 uses Lorentz symmetry, but at finite baryon density the system has a preferred frame. Whether the condensate shift gravitates as $w = -1$ or is absorbed into the $w = 0$ matter sector is the key physical question. The paper should engage with lattice QCD work on nucleon mass decomposition and gravitational form factors.

Response: This is the deepest physical question in the paper and the point where the derivation “from established physics” is most vulnerable. The argument from Lorentz scalar structure is necessary but may not be sufficient: at finite density, the condensate shift is a property of the medium, not of the vacuum in isolation. A full resolution requires computing the gravitational form factors of the nucleon with separate treatment of the scalar condensate component—a lattice QCD calculation that has not been performed. We acknowledge this as the principal open question of the programme and add it to §16. However, we note that the alternative (treating the $w = -1$ component as $w = 0$) is also an assumption—the standard one—and that our assumption at least has a symmetry argument in its favor.

Remaining Issue 2 (Numerical verification of Condition C is indirect): The simulations show convergent power spectra, not convergence of different initial conditions to the same fixed point.

Response: Accepted. This is a correct and important distinction. A direct test would run identical boxes with different random seeds and measure convergence of topological statistics. This has not been done and is added to §16 as a priority numerical test.

Remaining Issue 3 (“Full machinery of general relativity” overstated): The paper uses the Newtonian limit (Poisson equation), not the full Einstein equations.

Response: Accepted. The abstract is corrected to “the Newtonian limit of general relativity with a QCD-derived vacuum–matter coupling.”

Remaining Issue 4 (Corollary 11.4 should be downgraded): The Shannon–Kolmogorov approximation on which it rests is not strictly valid for the cosmological density field.

Response: Accepted. Corollary 11.4 is relabeled as an Observation (Remark) to reflect its dependence on the approximate relationship (35).